The raw memory coefficient in a central charge window Conserved shifts and the filtered linear field

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Abstract

We extend the project's second memory coefficient from exact half filling to a compact central charge window. The five raw quartic contractions acquire a finite charge-dependent correction entirely along the conserved quadratic polynomial. The linear rank filter has the same centered slope and limiting variance. Uniform rank tails and the existing charged lifts then transfer the complete conserved-subtracted two-energy spectrum. All finite-sector coefficients in growing counterterms are retained.

1 State and the central descendant table

Use standard CAR, the overlapping complex Gaussian p-body Hamiltonian, $\hbar = 1$, and the normalized maximally mixed trace in charge m. Norms mean $\|A\|_2^2 = \mathbb{E}\,\mathrm{tr}_m\, A^\dagger A$. Let $h = H_m$, $\mathcal{J}_h A = \{h, A\}/2$, $\mathcal{L}_h A = [h, A]$, and

$$N \text{ even}, \quad m = N/2 + xp, \quad |x| \le L, \quad \lambda_N = p^2/N \to \lambda > 0, \quad q_N = e^{-4\lambda_N}. \tag{1}$$

Integer admissible charges are understood; λ_N lies in a positive compact set. Put $C_m = \mathbb{E}\,\mathrm{tr}_m\, h^2 \to C = J_s^2/\lambda$, $B = h^2 - C_m I$, $A = \sum_i [H, c_i]^\dagger [H, c_i]$, $O = \mathsf{W}_2 A$. All W_j are Gaussian coupling chaos, not spectral Wick products. The output-deficit field $\mathcal{S}_N$ weights each quadratic product block with input ranks a, b and output rank r by $S = a + b - r$. With $DX = QX - \sum c_i^\dagger X c_i$ and $\mathcal{T}_j = (jp - D)H^j$ before restriction, define

$$\begin{aligned} E_0 &= \{A, h^2\}/2, & M_0 &= hAh, \\ S_0 &= \mathcal{J}_h \mathcal{T}_3 - E_0 - M_0, & C_0 &= \mathcal{T}_4 - 2E_0 - M_0 - 2S_0, \qquad Z_0 = h^4. \end{aligned} \tag{2}$$

For each row P, the central extension of the old table is

$$\mathsf{W}_2 P = C_m[p\ell_P + k_P + 2\lambda_N x \ell_P]B + C_m o_P O + j_P \mathcal{J}_h F + C_m s_P \mathcal{S}_N + o_{L^2}(1). \tag{3}$$

The old coefficients, listed on the next page, remain evaluated at finite λ_N, q_N. In particular the extra term is conserved; this statement concerns the central limit, not exact charge independence at finite N.

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2 Why no dynamic charge correction survives

P	ℓ_P	k_P	o_P	j_P	s_P
E_0	2	$-2\lambda q^2$	1	0	0
M_0	2	$-2\lambda(2q-1)$	q^2	0	0
S_0	$2q$	$-2\lambda(q^2-q+1)$	q	1	0
C_0	$2q^2$	$-2\lambda(8\lambda q^2-3q^2+2q+2)$	1	$2q$	$8\lambda q^2$
Z_0	0	q^2+2q+3	0	0	0

The exact two-charge contraction filters are those of the prior descendant calculation [2]. They use $n = N - m + 1$, two input ranks $a = p - u, b = p - v$, and output rank $r = 2p - u - v - S$. No charge-trace approximation is made in these filters. For covariance eigenvalue κ_j the central product expansion is

$$\frac{\kappa_{tp-d}}{C_m} = e^{-4\lambda_N t}\left[1 + \frac{4\lambda_N d - 4\lambda_N^2 t^2}{p} + O_{L,\lambda}\left(\frac{(1+d)^2}{p^2}\right)\right], \quad t = 1, 2. \tag{4}$$

It follows by summing the logarithms of $\prod_{j=0}^{p-1}(1-\nu/(m-j))(1-\nu/(N-m+p-j))$. The x terms cancel at order p^{-1} in this ratio. Substitution in the exact rational filters gives the five finite B shifts

$$4\lambda x\,(1, 1, q, q^2, 0) = 2\lambda x(\ell_E, \ell_M, \ell_S, \ell_C, \ell_Z). \tag{5}$$

All O, input-deficit and output-deficit coefficients agree with the half-filled values. For the raw core $\mathcal{E} = C_0 - 2S_0 + (E_0 + M_0)/2$, the net shift is $4\lambda x(1-q)^2 C_m B$, consistent with its independent filter.

Here are the norm controls promoting the expansions to (3). In the whole allowed rank domain the rational denominators have size bounded below by a constant times N or N^2. The A filters satisfy

$$|g_{A,P}/C_m - o_P| \le K(1 + u + v + S)/p, \qquad |g_{A,P}/C_m| \le K. \tag{6}$$

The established central weighted A norm and sixth output moment give a squared error at most $KC_m^4(p^{-2} + R^2/p^2 + p^4/R^6)$; set $R = p^{3/4}$. For the H^2 filter, subtraction of $p\ell_P$ leaves a bound $K(1+u+v+S)$. The factorial input tail and central sixth output moment give uniform integrability of its square. Equation (4) on fixed windows, followed by removal of the tails, therefore proves strong convergence. The conversion of its input-deficit part to $\mathcal{J}_h F$ is proved next.

3 The central linear filter and its covariance bridge

Write $h = \sum_{a=0}^{p} h_a$ with $\mathbb{E}\,\mathrm{tr}_m\, h_a^2 = C_m\omega_a$. The existing exact first-chaos coefficient t_a of $T_3 = \mathcal{T}_3 - 3\mathcal{J}_h A$ gives

$$F = \sum_a (t_a - \bar{t})h_a, \qquad \bar{t} = \sum_a \omega_a t_a. \tag{7}$$

For completeness set $k_a = \kappa_a/C_m$, $A_a = m - a$, $B_a = N - m - a + 1$, and $\alpha_a = a(N - a - p + 1)/(N - 2a + 1)$. The entire rank-dependent part of t_a/C_m is $k_a E_a$, where

$$E_a = (m-p)(1-p/A_a)(1+p/B_a) - m + 3[\alpha_a(1+p/B_a) + (p-\alpha_a)(1-p/A_a)]. \tag{8}$$

Thus the exact formula, rather than a trace transfer multiplied by p, gives for each fixed deficit u

$$(t_{p-u} - t_p)/C_m \to 8\lambda q u. \tag{9}$$

Uniformly on $0 \le a \le p$, $|E_a| \le Kp$, $|\partial_a E_a| \le K$, $0 \le k_a \le 1$, and $|\partial_a k_a| \le K/p$. Hence the left side of (9) is bounded by Ku over the entire rank range. The exact probability recurrence has ratio at most $K/(u+1)$ and tends to $\lambda/(u+1)$. Consequently $\omega_{p-u} \Rightarrow \mathrm{Pois}(\lambda)$ with uniform factorial tails, and centering is justified in every fixed polynomially weighted L^2 norm. In particular

$$F - 8\lambda_N q_N C_m (D_{\mathrm{in}} - \bar{u}h) \to 0 \quad (L^2), \tag{10}$$

$$\|F\|_2^2 \to 64\lambda^3 C^3 q^2 = 64 J_s^6 q^2, \tag{11}$$

$$\mathcal{J}_h F - 4\lambda_N q_N C_m L_{\mathrm{in}} \to 0 \quad (L^2). \tag{12}$$

Here $D_{\mathrm{in}} = \sum u h_{p-u}$, $\bar{u} = \sum u\omega_{p-u}$ and $L_{\mathrm{in}} = \mathsf{W}_2\{h, D_{\mathrm{in}} - \bar{u}h\}$. The final multiplication is controlled by Gaussian fourth moments: for an invariant linear Gaussian error X, $\mathbb{E}X^2 = vI$ and $\mathbb{E}\,\mathrm{tr}_m\, X^4 \le 3v^2$. Thus $\|hX\|_2 \le \sqrt{3C_m v}$.

The exact body-shift ratio is

$$R_j(u) = \frac{(u)_{\downarrow j}(N - 2p + u + 1)_{\uparrow j}(N - m + p - j + 1)_{\uparrow j}}{(m - p + 1)_{\uparrow j}[(p)_{\downarrow j}]^2} \longrightarrow \frac{(u)_{\downarrow j}}{\lambda^j}. \tag{13}$$

It has a uniform polynomial bound for fixed j. Thus the old signed $p, p-1, p-2$ covariance bridge and its whole-graph total-variation estimate apply uniformly in the central window. In particular $F/\|F\|_2$ is strongly approximated by $\sqrt{\lambda_N}\sum_a [R_1(p-a) - 1]h_a/\sqrt{C_m}$.

4 Charged marks and the full spectral interface

The lift of a $(p-1)$-body field U is $g(Q)U$, with $d = m - p + 1$ and $g(n) = (n - p + 1)/d$. The reduced-rank binomial identity and the direct CAR identities are valid at general charge:

$$\mathcal{M}_+(gU, V) = (1 + d^{-1})\mathcal{M}_+(U, V) - d^{-1}U\overline{D}V, \tag{14}$$

$$\mathcal{M}_-(gU, V) = (1 - d^{-1})\mathcal{M}_-(U, V) + d^{-1}UDV, \tag{15}$$

$$[(kp - D)(gB')]_m = (1 - d^{-1})[(kp - 1 - D)B']_m + (k - 1)pB'_m/d. \tag{16}$$

Here $\mathcal{M}_- = -\sum[U, c_i^\dagger][V, c_i]$, $\mathcal{M}_+ = -\sum[U, c_i][V, c_i^\dagger]$, and B' contains one shifted slot in a fixed-length neutral product. Reversing the order in (15) puts DV or $\overline{D}V$ on the left. These formulas were already derived and checked in the mixed-spectrum note [3]; no new lift is postulated.

The preceding central Casimir identity gives, for a p-body slot,

$$DV = pV, \qquad \overline{D}V = pV - 2D_{\mathrm{in}}V - E_{\mathrm{in}}V, \qquad (E_{\mathrm{in}}V)_a = \frac{u(u - 1 + N - 2m)}{m - p + 1}V_a. \tag{17}$$

Factorial tails bound the input terms in every fixed moment. Since $N - 2m = -2xp$, the last term is $O(p^{-1})$, and all lift corrections are $O(p/N) = O(p^{-1})$ in the fixed-degree mixed traces. The quartic identity also has this size.

The existing oriented chord rule permits unequal index-set sizes [1]; changing p to $p-1$ preserves its limiting overlap parameters. The central marked-graph trace theorem and exponential overlap bounds transfer its fixed derivatives. Every graph in a cross with one F therefore has two signed terms with the same limit, and cancels. The lower/upper force and quartic lifts cover F–Y_-, F–Y_+, F–R and F–Q_4. The existing endpoint certificates transfer the remaining crosses. In applying them, one first combines the raw words into bounded marked Wick blocks; one never multiplies an $O(p^{-1})$ trace error by an unremoved p counterterm.

We obtain the same diagonal two-energy basis (Y, F, R, T, Q_4) as at half filling, with parameters q, C. In particular

$$d\nu_F = 64\lambda^3 C^3 q^2\, d\mu_q, \qquad \nu_{F,Y_\pm} = \nu_{F,R} = \nu_{F,Q_4} = 0. \tag{18}$$

For marked representatives these follow from all fixed energy moments, tightness, and compact moment uniqueness. Strong replacements transfer bounded spectral tests by Cauchy–Schwarz. Energy-weighted uses additionally employ the factorial weighted errors and fixed-degree Gaussian/marked moment bounds above. Ordinary chaos orthogonality alone would not give these statements.

5 The conserved subtraction and restored raw coefficient

Put $d_3 = q^2 - q + 1$, $d_5 = 1 + q + q^2 + q^3 + q^4$, and

$$\alpha = -2q^3/(d_3 d_5), \qquad \beta = -q^2(1 + q + q^2)/(d_3 d_5), \qquad \gamma = 2q/d_3,$$
$$\delta = \frac{2\lambda(1-q)^3(1+q)(1+q+q^2)}{(1+q^2)d_5}, \qquad g_J = \alpha + \beta + \gamma - 1. \tag{19}$$

Let $\mathcal{Q} = C_0 - \alpha E_0 - \beta M_0 - \gamma S_0 - \delta Z_0$, $Q_4 = \mathsf{W}_4\mathcal{Q}$, $e_{\mathcal{Q}} = \mathbb{E}\,\mathrm{tr}_m\,\mathcal{Q}$. For any table column set $f_Q = f_C - \alpha f_E - \beta f_M - \gamma f_S - \delta f_Z$. Its central correction is exactly $2\lambda_N x \ell_Q C_m B$ to the order under consideration.

Use actual finite-sector $a_m = \mathbb{E}\,\mathrm{tr}_m\,A$, $c_m = \langle O, B\rangle/\langle B, B\rangle$, $K_A = a_m I + c_m B$, $Y = A - K_A$. Define

$$a = -(1-q)^2/(1+q^2), \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)},$$
$$P_m = (ac_m + b)h^3 + [\bar{t} - (ac_m + b)\bar{z}]h, \qquad \bar{z} = \langle h, \mathsf{W}_1 h^3\rangle/C_m. \tag{20}$$

The exact cubic reduction has the same central strong remainder: $T_3 - P_m = R + a\mathcal{J}_h Y + F + o_{L^2}(1)$. Here $R = \mathsf{W}_3(T_3 - a\mathcal{J}_h A - bh^3)$. Indeed $B_Z, B_S = O_{L^2}(p^{-1})$ follow from the centered k_a and $k_a J_a$ slopes, as in the preceding cubic calculation. Their limits are $pB_Z - F/2 \to 0$ and $pB_S - (1-2\lambda)F/2 \to 0$, also with fixed energy weights.

With $r_N = p/m$, $c = 2\lambda(1-q)/(1+q)$, set

$$D_{2,m} = 2r_N h P_m + 3r_N^2 h^2 K_A - K_A^2/m - m r_N^4 h^4,$$
$$\mathcal{C}_{2,m} = D_{2,m} + e_{\mathcal{Q}} I + \delta h^4 + g_J h^2 K_A + b_R h P_m$$
$$+ C_m[p\ell_Q + k_Q + 2\lambda_N x \ell_Q + o_Q c_m + 8\lambda_N q_N^2 c]B. \tag{21}$$

Every term commutes with h sample by sample. All q, λ in this finite-N definition mean q_N, λ_N; $C_m, c_m, \bar{t}, \bar{z}, e_{\mathcal{Q}}$ remain exact sector quantities. The renewal identity and (3) give

$$U_{2,m} := mZ_2 - \mathcal{C}_{2,m} = X + b_R\mathcal{J}_h R + b_F\mathcal{J}_h F + b_J\mathcal{J}_h^2 Y + b_D\mathcal{L}_h^2 Y + C\kappa Y + o_{L^2}(1), \tag{22}$$

Here $\widehat{M}(s) = G + isZ_1 - s^2 Z_2/2 + O(s^3)$ defines the finite-N centered memory coefficients. Also $T = (1 - \Pi_N)\mathcal{S}_N$ with $\Pi_N = \mathrm{proj}_{\{B, L_{\mathrm{in}}, O\}}$, $X = Q_4 + 8\lambda C q^2 T$, and

$$b_R = -2(1-q)^2/d_3, \qquad b_F = 2(q-1), \qquad b_D = q^2/(4d_5),$$
$$b_J = g_J + ab_R, \qquad \kappa = (1-q)(1+q)^2/d_5 + 8\lambda q^2\rho, \tag{23}$$

$\rho = v/(v + \Delta)$, $v = 2\lambda(1-q) + 16\lambda^2 q/(1+q)$, $\Delta = 2\lambda(1+q)$. The renewal error is $O_{L^2}(p^{-1})$: its terms have factors r_N, r_N^2, or $a_m/m = O(p^{-1})$ and bounded centered higher moments. The finite conserved K_A^2/m term is retained in (21).

6 Spectrum, evidence, and continuation

Let $e = (E + E')/2$, $\omega = E - E'$, $H_2 = b_J e^2 + b_D \omega^2 + C\kappa$. The result is

$$d\nu_{U_2} = V_4 d\mu_{q^4} + V_2 d\mu_{q^2} + b_R^2 e^2 V_R d\mu_{q^3} + b_F^2 e^2 V_F d\mu_q + W H_2^2 d\mu_{q^2}. \tag{24}$$

These are the same explicit weights as in the raw-spectrum note [4], with $W = 4C^2\lambda + 16C^2\lambda^2 q/(1+q)$, V_F in (11), and $V_2 = 64\lambda^2 C^4 q^4 v\Delta/(v+\Delta)$. The existing Mehler measure μ_r has variance-C marginals. The cross measures are

$$d\nu_{Y,U_2} = W H_2 d\mu_{q^2}, \qquad d\nu_{L,U_2} = aeW H_2 d\mu_{q^2} + b_F e V_F d\mu_q + b_R e V_R d\mu_{q^3}. \tag{25}$$

For densities $w = WK_2$, $f = V_F K_1$, $r = V_R K_3$, $z = V_4 K_4 + V_2 K_2$, the (Y, L, U_2) matrix is

$$M \operatorname{diag}(w, f, r, z) M^T, \qquad M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ ae & 1 & 1 & 0 \\ H_2 & b_F e & b_R e & 1 \end{pmatrix}. \tag{26}$$

Its determinant is $w[(f+r)z + e^2 fr(b_F - b_R)^2]$. Thus the compact central window changes the required subtraction, but not the limiting dynamic spectral matrix. The ordered small-λ shape already derived in Ref. [4] is inherited after this central double-scaled limit.

Executed evidence. The restored symbolic certificate checks all five central filters, the cubic slope, and the one-body-shift ratio. Its recorded source hashes were verified before reuse. A new calculation checks 123 exact rational coefficient references and 36 full-rank sums at $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, $x = -2, 0, 2$, each at 80 and 120 decimal digits. All $p+1$ ranks are included; finite variance differences, signed-covariance errors, weighted tails, and both cubic remainders are saved. Agreement of numerical precisions is not a convergence proof or an interval certificate. The assembly verifies seven prior/current certificates and the positive matrix algebra, without repeating old endpoint scans.

Next calculation. The next issue is charge summation: obtain bounds outside every fixed central window for the *conserved-subtracted* fields, then combine them with the binomial charge law. Compact-x uniformity alone cannot interchange that sum with the limit. Afterwards control the two-insertion finite-memory remainder and connect the bilocal to the Susskind bulk meson dictionary. This note identifies fixed Taylor coefficients, not the finite-memory kernel.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sec. 3.3.

[2] A. Day and Chengheng, local notes: *Restoring the raw memory descendants*; *Central coherent norms*; *Central force spectrum* (2026).

[3] A. Day and Chengheng, local notes: *Mixed spectra of the first memory fields*; *Linear rank-filter covariance bridge* (2026).

[4] A. Day and Chengheng, *The second coefficient after a conserved subtraction*, local six-page note, 6 November 2026.